CASE STUDY 04

Employee Payroll and Deduction Analysis

1. PROBLEM STATEMENT

Calculate an employee's gross salary from basic pay, allowances, overtime, and incentives. Apply deductions and print a

salary statement.

2. OBJECTIVES

• Calculate allowances and overtime earnings.

• Find gross salary and total deductions.

• Display net salary in a readable format.

3. ASSUMPTIONS

• Dearness allowance (DA) is 10% of basic pay.

• House rent allowance (HRA) is 20% of basic pay.

• Overtime is paid at ■200 per hour.

• Professional deduction is ■200.

• Income deduction is 5% of gross salary when gross salary exceeds ■30,000; otherwise it is 2%.

4. ALGORITHM

1. Read employee name, basic pay, overtime hours and incentive.

2. Calculate DA, HRA and overtime pay.

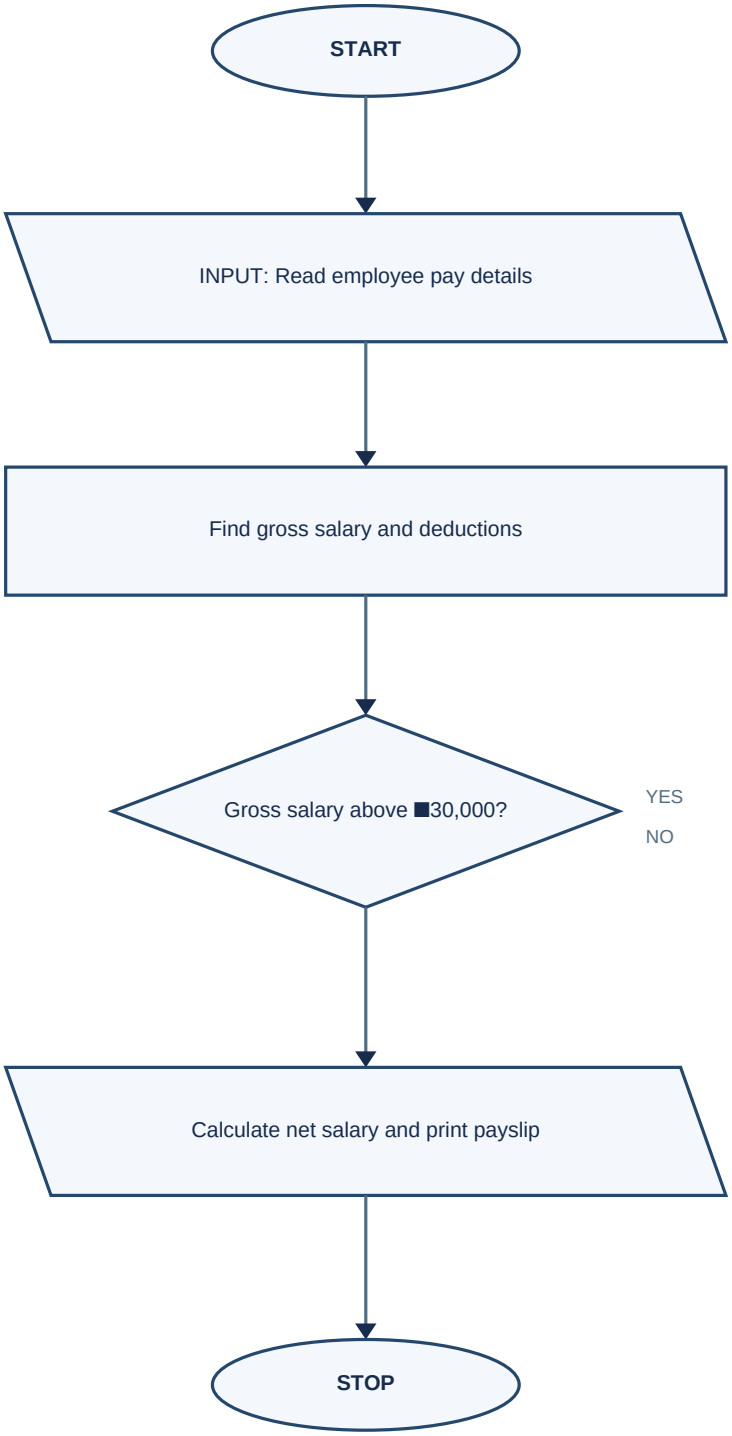
3. Add all earnings to obtain gross salary.

4. Select the deduction rate based on whether gross salary exceeds ■30,000; calculate total deductions.

5. Subtract deductions from gross salary and display the payslip.

5. FLOWCHART

Oval: START/STOP | Parallelogram: input/output | Rectangle: process/calculation | Diamond: decision



6. PSEUDOCODE

```
BEGIN
INPUT name, basic, overtime_hours, incentive
DA = basic * 0.10
HRA = basic * 0.20
overtime_pay = overtime_hours * 200
gross = basic + DA + HRA + overtime_pay + incentive
IF gross > 30000: rate = 0.05
ELSE: rate = 0.02
CALCULATE income and professional deductions
net_salary = gross - total_deduction
DISPLAY formatted employee payslip
END
```

7. PYTHON SOURCE CODE

```
name = "Ravi"
basic = 25000
overtime_hours = 5
incentive = 2000
da = basic * 0.10
hra = basic * 0.20
overtime_pay = overtime_hours * 200
gross = basic + da + hra + overtime_pay + incentive
if gross > 30000:
deduction_rate = 0.05
else:
deduction_rate = 0.02
income_deduction = gross * deduction_rate
professional_deduction = 200
total_deduction = income_deduction + professional_deduction
net_salary = gross - total_deduction
print("EMPLOYEE PAYSLIP")
print("Name:", name)
print("Basic Pay: Rs.", basic)
print("DA: Rs.", da)
print("HRA: Rs.", hra)
print("Overtime: Rs.", overtime_pay)
print("Incentive: Rs.", incentive)
print("Gross Salary: Rs.", gross)
print("Total Deduction: Rs.", total_deduction)
print("Net Salary: Rs.", net_salary)
```

8. TEST CASES & EXPECTED RESULTS

TEST INPUT / CONDITION REASONING EXPECTED RESULT

Basic ■25000, overtime 5 hours, DA ■2500; HRA ■5000; overtime ■1000 Gross ■35500; net ■33525

incentive ■2000

Basic ■20000, overtime 0, incentive Gross ■26000; 2% plus ■200 deduction Net ■25280

■0

9. SAMPLE OUTPUT

```
EMPLOYEE PAYSLIP
Name: Ravi
Basic Pay: Rs. 25000
DA: Rs. 2500.0
HRA: Rs. 5000.0
Overtime: Rs. 1000
Incentive: Rs. 2000
Gross Salary: Rs. 35500.0
Total Deduction: Rs. 1975.0
Net Salary: Rs. 33525.0
```

10. CONCLUSION

The payroll program combines different salary components, applies the specified deductions, and produces a clear net

salary statement.

SYLLABUS CONCEPTS Variables • Data Types • Operators • Conditions • Loops • Lists • Nested Lists • Tuples • String Operations